

D7054E Project Assignment

Abdalrhman Dabor

Group 16

March 12, 2026

1 Introduction

The dataset that we are going to use contains historical video game sales data collected from 1980 to 2016. It includes over 16,000 video game records across multiple platforms and regions. Each record includes information such as the game's platform, genre, publisher, release year, regional sales, and total global sales. The data will be retrieved by downloading the CSV file directly from Kaggle, which can be found here [\[1\]](#)

1.1 What is the main problem addressed?

It helps video game companies estimate the expected global sales of a new game based on its attributes and whether it will be a high-seller. This helps to make decisions before developing a game, and can give insights into expected sales and market performance. By learning from previous games' data, we can reduce financial risks by identifying high-potential projects and optimizing resource allocation.

1.2 Hypothesis

1. We hypothesize that games in the Action and Sports genres achieve significantly higher global sales compared to genres such as Puzzle or Adventure.
2. We hypothesize that platforms such as PS2, PS3, and Xbox 360 were the most successful in terms of sales during the period 2000–2010, while PC platform became more important after 2010.

2 Method

The dataset provides a suitable basis for analyzing sales trends and building predictive models to estimate the global commercial performance of video games. The data will be retrieved by downloading the CSV file directly from Kaggle and loaded using the Python pandas library. Figure 1 presents the overall block diagram of the data science pipeline used in this project.

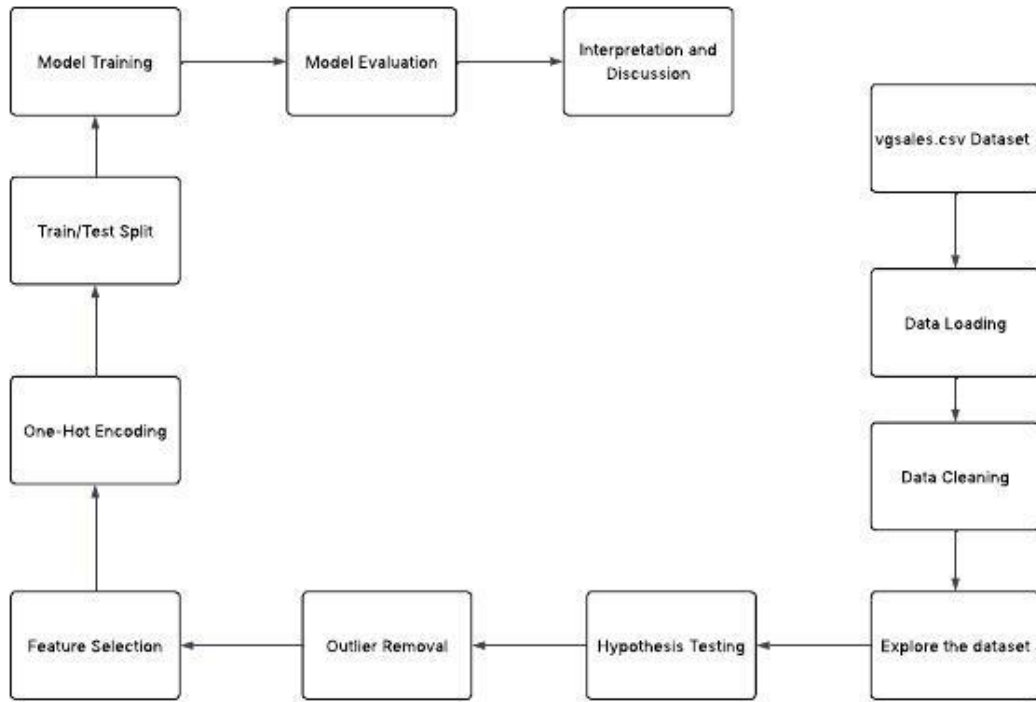


Figure 1: Block Diagram of the Data Science Pipeline used in the project

Here is the pipeline of how the data was processed:

- Data Cleaning & Outlier Removal:** First, we removed any rows with missing values and all rows with unrealistic year values to ensure the models work correctly. Number of rows before cleaning = 16598, and Number of rows after cleaning = 16291. So, Total rows removed = 307. The code snippet in figure 2 shows the main cleaning steps used in the project: loading the dataset, removing missing values, converting the Year column into integers, and filtering unrealistic year values. Moreover, during data exploration, we found that a few games had massive sales (over 10 million copies). We filtered the data to remove games that sold more than 10 million copies, as these are rare cases and do not represent the typical market (62 games out of 16287). We observed that the model's accuracy improved after removing these outliers.

```

# ===== Clean the data =====
# Load the dataset
df = pd.read_csv('vgsales.csv')

# Just to see the number of rows BEFORE cleaning
rows_before = len(df)
print(f"Number of rows before cleaning: {rows_before}")

# Remove any rows that have missing values
df_clean = df.dropna().copy()

# Convert Year column from float to integer
df_clean['Year'] = df_clean['Year'].astype(int)

# Remove any rows with unrealistic year values
df_clean = df_clean[df_clean['Year'] <= 2016]

|

# Just to see the number of rows AFTER cleaning
rows_after = len(df_clean)
print(f"Number of rows after cleaning: {rows_after}")
# Calculate how many rows were removed
print(f"Total rows removed: {rows_before - rows_after}")

```

Figure 2: Data Cleaning Code Snippet

- **Feature Extraction:** Categorical features such as Genre, Platform, and Publisher were converted into numerical form using one-hot encoding so the machine learning models can use them. The Year feature was converted into an integer since it was float in the original data. The snippet in figure 3 shows how the selected input features were prepared and how categorical variables were converted into numerical form using one-hot encoding.

```
# ===== Prepare features and target =====

# Just to see Total games with over 10 million sales
outliers = df_clean[df_clean['Global_Sales'] > 10]
print(f"Total games with over 10 million sales: {len(outliers)}")

# Keep only games with 10 million sales or fewer
df_no_outliers = df_clean[df_clean['Global_Sales'] <= 10]

# Select the features (inputs) and the target (output)
# Inputs
X = df_no_outliers[['Platform', 'Genre', 'Publisher', 'Year']]
# output
y = df_no_outliers['Global_Sales']

# Convert text columns to numbers(0&1) using One-Hot Encoding so the model reads them
X_encoded = pd.get_dummies(X, columns=['Platform', 'Genre', 'Publisher'])

sns.histplot(df_no_outliers['Global_Sales'], bins=30)
plt.title("Distribution of Global Video Game Sales")
plt.xlabel("Global Sales (Millions)")
plt.ylabel("Frequency")
plt.show()
```

Figure 3: One-hot Encoding Code Snippet

- **Algorithms:** The dataset was split into training (80%) and testing (20%) sets to evaluate model performance. We used simple supervised regression models. We chose Linear Regression as the baseline model to predict Global Sales. Moreover, Decision Tree Regressor was also used to capture non-linear relationships. The snippet in figure 4 presents the training and evaluation process for the Linear Regression model using Mean Squared Error and the R^2 score.

```
# ===== Model Training =====
# 80% of the data will be used for TRAINING
# 20% of the data will be used for TESTING
X_train, X_test, y_train, y_test = train_test_split(X_encoded, y, test_size=0.2, random_state=42)

# Train the model using Linear Regression
lr_model = LinearRegression()
lr_model.fit(X_train, y_train)

# Train the model using Decision Tree Regressor
dt_model = DecisionTreeRegressor(max_depth=10, random_state=42)
dt_model.fit(X_train, y_train)
```

Figure 4: Training Code Snippet

3 Results

3.1 Hypothesis Testing

We hypothesize that games in the Action and Sports genres achieve significantly higher global sales compared to genres such as Puzzle or Adventure

To test this hypothesis, we filtered the dataset to include only "Action", "Sports", "Puzzle", and "Adventure" games. We created a bar plot using seaborn and calculated the exact mean sales for each genre. Graph 5 and the printed data

6 clearly show that Sports and Action games have higher average sales compared to Puzzle and Adventure games, which supports the first hypothesis.

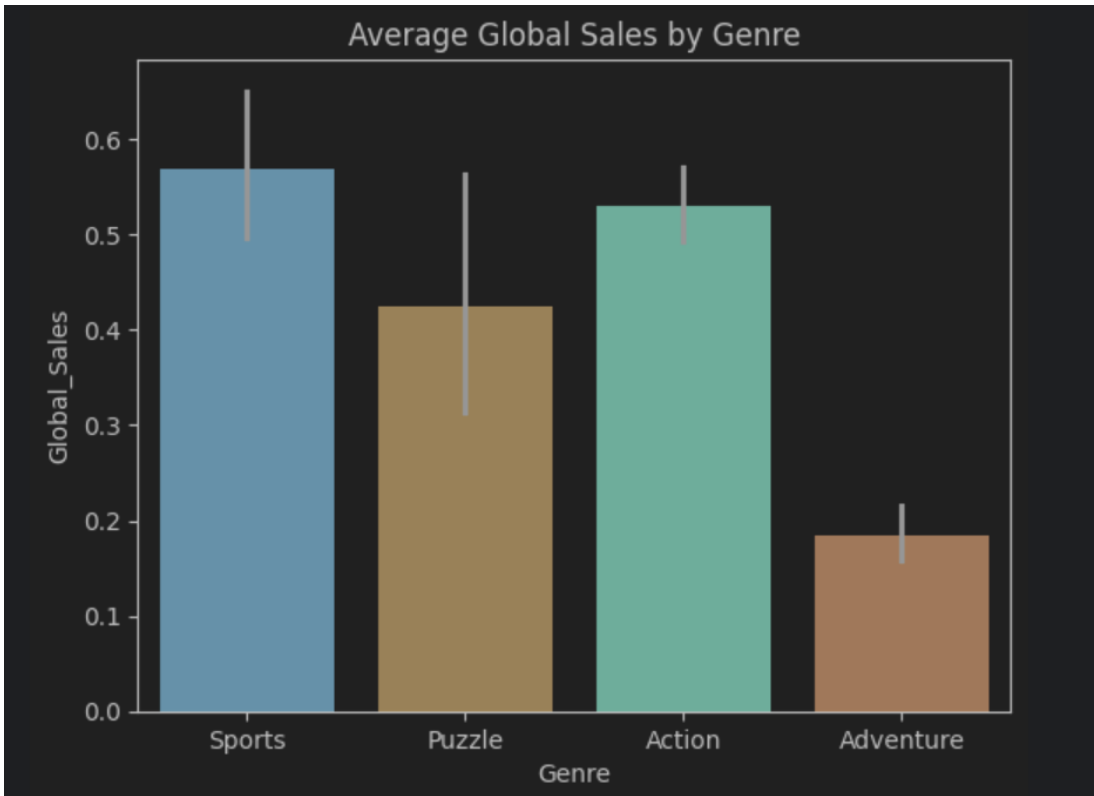


Figure 5: Average Global Sales by Genre (Graph)

	Genre	Global_Sales
3	Sports	0.568247
0	Action	0.530102
2	Puzzle	0.424930
1	Adventure	0.184137

Figure 6: Average Global Sales by Genre (Numerical Results)

We hypothesize that platforms such as PS2, PS3, and Xbox 360 were the most successful in terms of sales during the period 2000–2010, while PC platform became more important after 2010

To test this hypothesis, we split the data into two periods:2000-2010 and post-2010
For the first period (2000-2010): The results matched the hypothesis for the period. PS2, Xbox 360, and PS3 were among the most successful platforms at that time frame as shown in the graph 7.
For the second period (post-2010): The PC platform maintained a strong and consistent presence in the market, proving its long-term relevance. However, the visualizations also highlight that newer generations of traditional

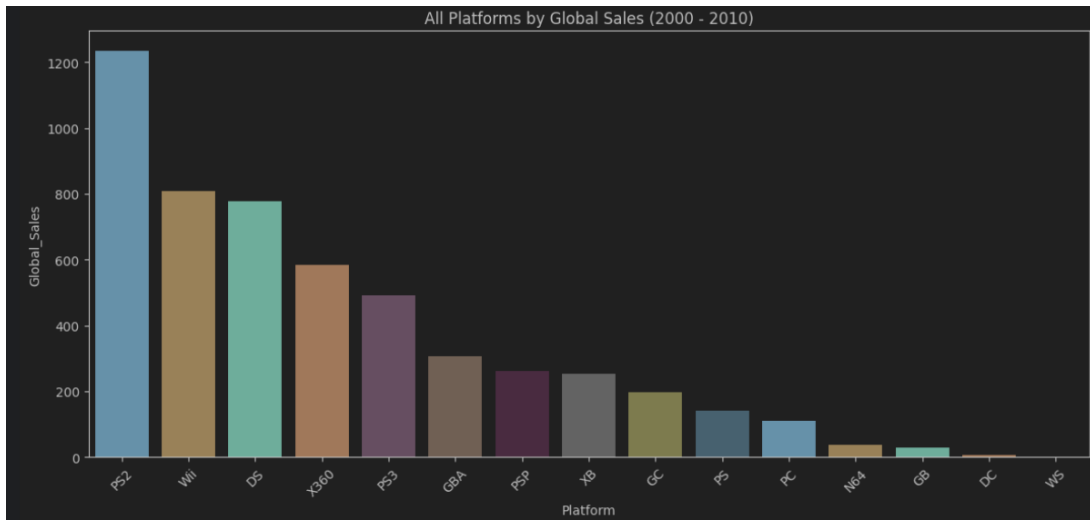


Figure 7: All Platforms by Global Sales (2000 - 2010)

consoles, specifically the PS4 and Xbox One, continued to lead overall top sales alongside the PC as shown in the graph 8.

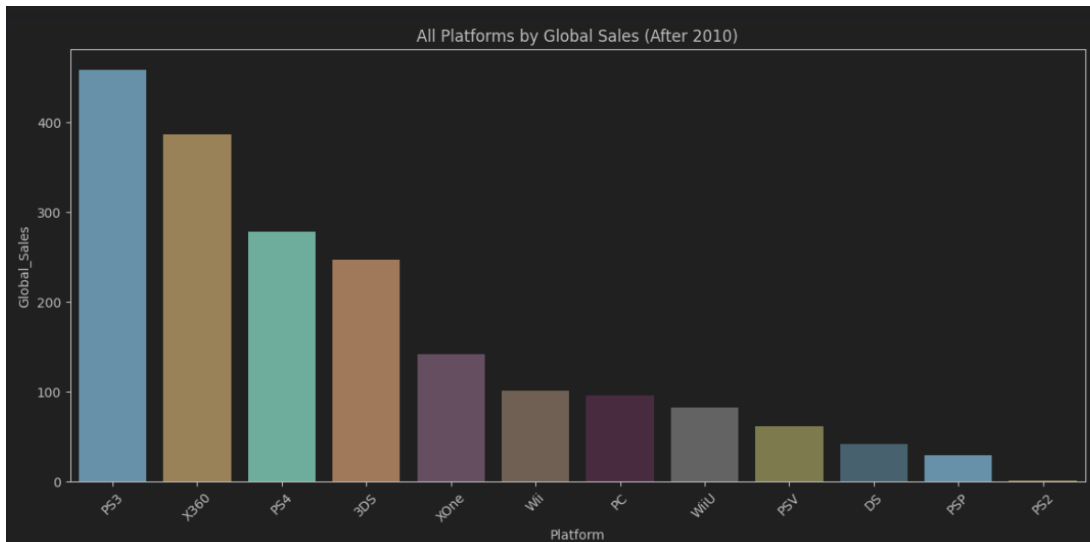


Figure 8: All Platforms by Global Sales (After 2010)

Therefore, the second hypothesis is only partially supported.

Data Distribution

Before training the models, I plotted a histogram to understand how the sales data is distributed across the games after removing the extreme outliers (games over 10 million sales). The histogram 9 shows that the vast majority of games sell at the lower end of the spectrum (between 0 and 2 million copies), with the frequency dropping significantly as sales approach the 10 million limit.

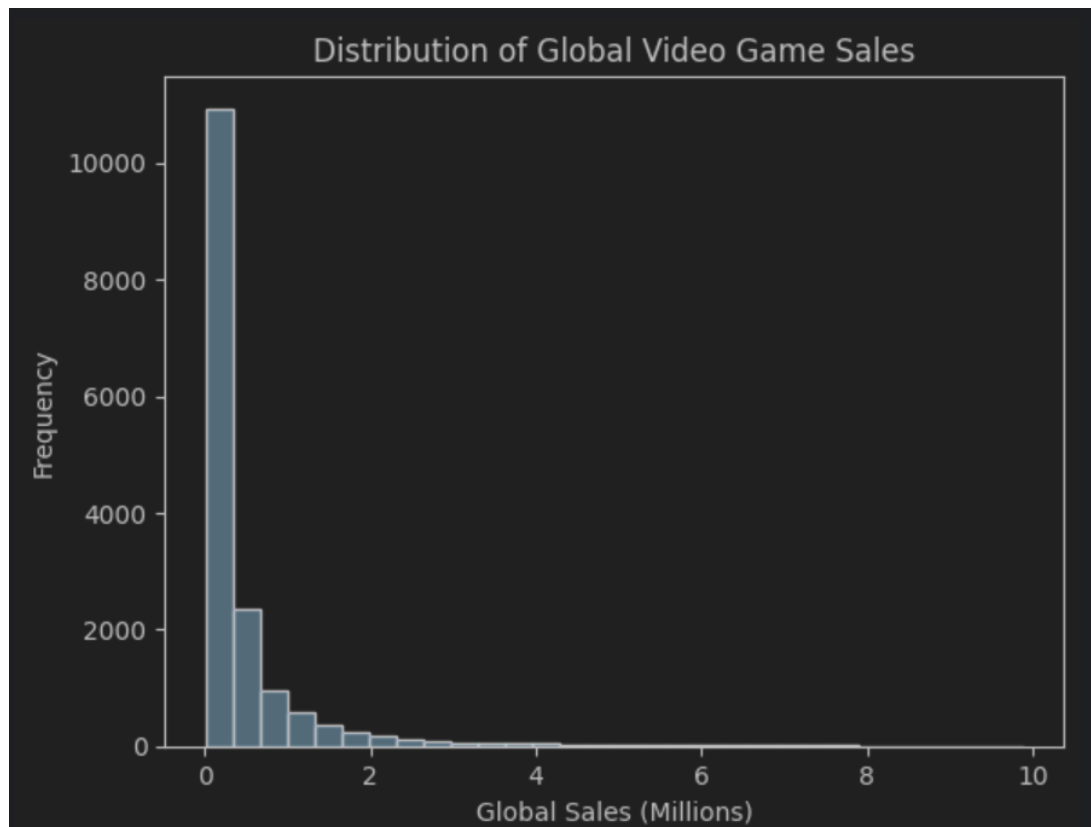


Figure 9: Distribution of Global Video Game Sales

Model Performance

Model evaluation used Mean Squared Error and R2 score. After training both models on the cleaned dataset, we got the following results on the testing data:

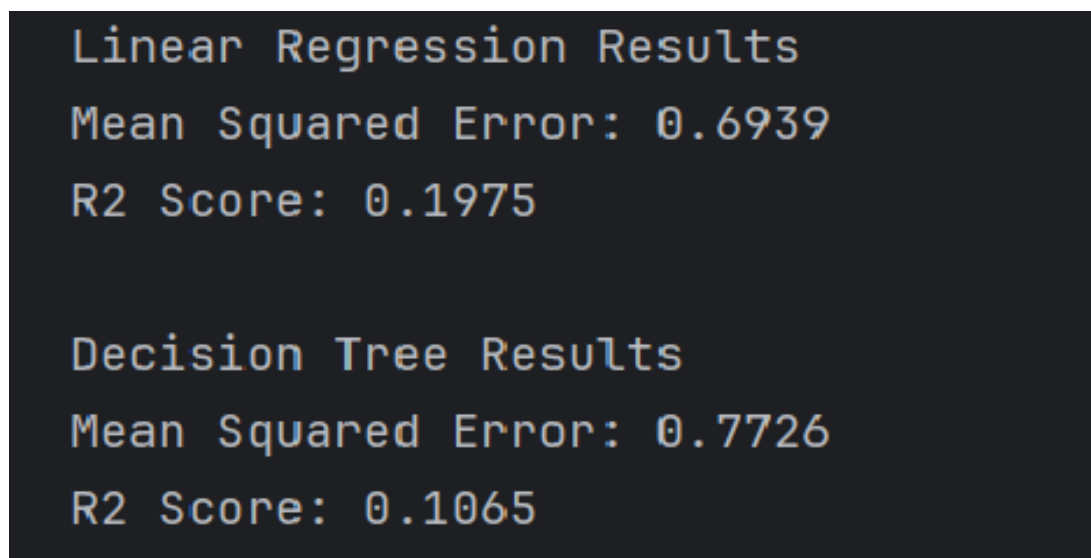


Figure 10: Model evaluation

To better visualize how the Linear Regression model performed, I created a scatter plot comparing the actual sales versus the predicted sales against a perfect prediction line (in red).

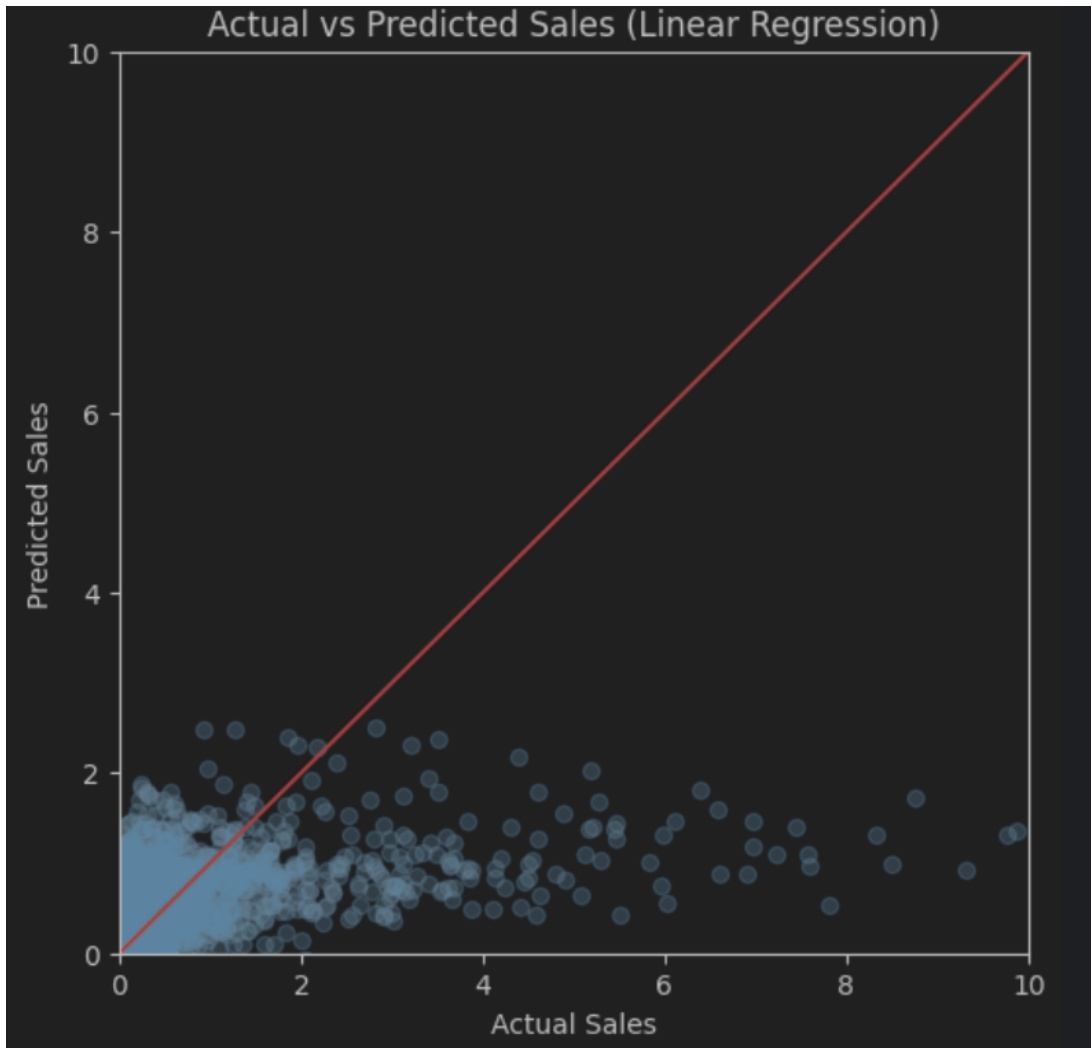


Figure 11: Actual vs Predicted Sales (Linear Regression)

4 Discussion

The Linear Regression model was much better than the Decision Tree. By removing the extreme outliers, the Linear Regression's Mean Squared Error dropped to just 0.69. This means when the model guesses the sales of a normal game, it is usually off by less than 1 million copies, which is a very good and realistic margin for the gaming industry.

The scatter plot visualizes this well; while the model captures the general trend at lower sales volumes, it struggles slightly as sales get higher, which is typical for such unpredictable markets.

The R^2 score for Linear Regression was about 20% (0.1975). This means that factors like the platform, genre, publisher, and year explain 20% of why a game sells well. While 20% might seem low, it is logical. Game sales rely heavily on other critical factors not included in this dataset, such as the marketing budget, graphics quality, and most importantly, player ratings.

For future work, extracting actual player ratings from platforms like the official Quest store or Steam would likely increase the model's accuracy significantly, as user reviews drive modern game sales.

The Decision Tree performed poorly (only 10.65% R^2 score), showing that the relationships in this data are more linear than hierarchical.

4.1 Data Ethics

When using predictive models like this, there are ethical considerations. There is a chance that the model leads to Bias toward one genre of games (like Sports or Action) to be always profitable, so all publishers invest in this genre only. This could kill creativity and variety in the gaming market. Also, it must be ensured that the data used does not infringe on the rights of publishing companies.

4.2 Libraries and tools used

- pandas for data manipulation.
- matplotlib.pyplot and seaborn for data visualization.
- scikit-learn for splitting data, building the machine learning models, and calculating evaluation metrics.

References

- [1] Video Game Sales Dataset. Available at: <https://www.kaggle.com/datasets/gregorut/videogamesales>